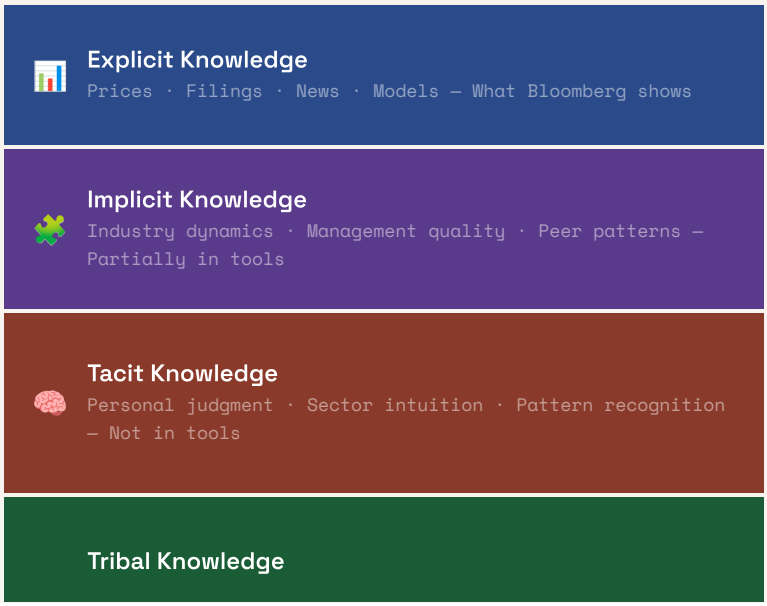
TRIBAL KNOWLEDGE GAP ANALYSIS · FEBRUARY 2026

What the tools show. What the *expert knows.* The gap between.

A veteran portfolio manager and a junior analyst looking at the same Bloomberg terminal see entirely different things. The terminal is identical. The knowledge they bring to it is not. That gap — between what tools provide and what experienced humans carry in their heads — is precisely where most

THE KNOWLEDGE ICEBERG



investment edge actually lives. And it's almost entirely unencoded.

🗣 Undocumented relationship maps · Shared desk heuristics · "Everyone knows X but it's not written anywhere" — Completely outside tools

42% of institutional knowledge lives only in individual heads — Panopto Research

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SECTION 01 · THE FOUR KNOWLEDGE LAYERS

What tools give. What humans carry. *Where the edge actually lives.*

There are four distinct layers of knowledge a portfolio manager brings to every investment decision. Commercial tools serve the top layer brilliantly and the second partially. The bottom two — where the genuine edge lives — are completely unencoded.

📊 WELL-SERVED

Explicit

DOCUMENTED · TRANSFERABLE · QUERYABLE

What is written down. Prices, financial statements, earnings releases, SEC filings, analyst reports, news articles. Unambiguous. Any analyst can retrieve it instantly. Zero interpretation required. Zero edge remaining.

Examples:

🧩 PARTIALLY SERVED

Implicit

INFERRABLE · SLOW · ANALYST-DEPENDENT

Knowledge that can be extracted from explicit data with effort. Industry dynamics, management tone quality on earnings calls, peer relationship patterns, sector cycle timing. Requires interpretation. Edge varies by analyst skill.

🧠 NOT IN TOOLS

Tacit

PERSONAL · EXPERIENTIAL · IMPOSSIBLE TO DOCUMENT

Personal intuition built from years of experience. The "feel" for when a CEO is nervous, when a sector is about to inflect, when a price pattern looks familiar. Kahneman's System 1 thinking applied to markets. Not transferable. Dies when the

🗣 COMPLETELY UNENCODED

Tribal

SHARED · UNWRITTEN · GROUP-SPECIFIC · THE REAL EDGE

Shared, undocumented knowledge held collectively by a team or desk. "We all know" that these two companies move together. "Everyone here knows" this CEO calls this specific sell-side analyst before announcements. "It's common

TSMC Q4 2025 revenue: $26.9B (+37% YoY). NVIDIA stock price: $145.83 at close. S&P 500 PE ratio: 28.6x.

TOOLS THAT COVER IT

Bloomberg
FactSet
Capital IQ
LSEG

Partially served by AI search tools.

Examples:

"TSMC's management sounded more confident than usual on pricing power." "NVIDIA's margins are expanding faster than Street consensus."

TOOLS THAT COVER IT

AlphaSense
RavenPack
Tegus
Partially

analyst leaves. Zero tool coverage.

Examples:

"I've seen this pattern in semis before — this is the early phase of an upcycle." "Something felt off about that CFO's tone when asked about Q3 guidance."

TOOLS THAT COVER IT

None
Essentia Analytics (partial)

knowledge on the desk" that this supply chain link is underdisclosed. Passed down verbally. Lost with turnover.

Examples:

"Everyone on this desk knows AMAT moves 3 weeks after ASML." "This substrate supplier is always the canary — never covered by anyone."

TOOLS THAT COVER IT

None exist
MKG (target)

"42% of institutional knowledge is unique to individual employees and isn't documented anywhere else in the organization. In investment management, where the 'institution' is a fund's accumulated relationship map, this figure is almost certainly higher."

PANOPTO RESEARCH · KNOWLEDGE RETENTION STUDY · APPLIED TO INVESTMENT MANAGEMENT CONTEXT

SECTION 02 · TAXONOMY OF FINANCIAL TRIBAL KNOWLEDGE

The six things a veteran PM *knows that no tool shows*

These are not vague intuitions. They are specific, actionable, computable insights — carried verbally in experienced heads, lost on departure, and completely invisible to every commercial financial data platform.

01

The Secret Relationship Map — "These two move together, nobody covers it"

Relationship

Every seasoned sector analyst carries a mental map of undisclosed or under-covered relationship edges. These are not in Capital IQ. They don't appear in Bloomberg relationship databases. They emerge from years of reading earnings call footnotes, noticing pattern co-movement, attending industry conferences, and reading regional press that no data vendor indexes. A veteran semiconductor analyst knows 50 relationships that the database doesn't.

> *"AMAT always moves 2–3 weeks after ASML results, because their booking lead times are almost identical but AMAT's IR doesn't front-run it the way ASML's does. It's not in any database. You just know it after two cycles."*

The problem is structural: Capital IQ records relationships from filings. But the most valuable relationship intelligence comes from *patterns not disclosed in filings* — operational dependencies revealed only in aggregate market behavior, industry contacts, and historical co-movement. When the analyst who tracked this leaves, the knowledge goes with them.

MKG BRIDGE POINT

Graph construction from co-movement signals + earnings call NLP extracts the implicit relationship structure. Edge weights built from co-mention frequency, lead-lag co-movement, and supply chain filing cross-reference surface relationships that are "known but unwritten" — encoding tribal knowledge into graph edges that survive analyst turnover.

02 Company-Specific Calibration — "This CEO always sandbagged by 15%"

Calibration

Experienced analysts develop precise, company-specific calibration adjustments that are never written in any research note. They know which CEOs systematically underpromise and which overstate. They know which quarters of the year a company consistently surprises in one direction. They know which management teams get nervous under pressure from specific analysts, and what that nervousness signals. Fuller & Thaler built a $3B fund exploiting anchoring bias in analyst estimates — exactly this type of calibration. But it lives entirely in human heads at most funds.

> *"This CFO always guides down 8–12% from what they actually expect in Q3. It's his style. We adjust for it every single time. It's never written in any note — it's just what the desk knows."*

MKG BRIDGE POINT

Historical guidance vs. actuals database per entity per executive. WAN learns systematic over/under-guidance patterns per management team from historical earnings data. Calibration factors become explicit, computable, and survivable beyond individual analyst tenure.

03 The Canary Companies — "Watch this one, nobody else does"

Leading Indicator

In every sector, there are small, under-covered companies whose results systematically lead larger, better-covered companies by weeks. Experienced sector analysts know these canaries intimately. They watch small substrate suppliers before semiconductors, small trucking companies before industrials, obscure chemicals distributors before energy. These are not discoverable by searching Bloomberg. They are found through years of sector immersion and pattern recognition. The research finding from Shanghai lockdown data confirms it: tier-1 suppliers showed significant abnormal returns before focal firms did. The canaries exist. Knowing which ones requires tribal knowledge.

> *"There's a substrate supplier in Taiwan — 10 analysts cover it globally. When their order intake drops more than 12%, TSMC's advanced packaging revenue misses 6 weeks later. Every cycle, like clockwork. It's not in any database."*

MKG BRIDGE POINT

Lead-lag relationship detection across the graph: edges with consistent temporal precedence (Company A moves before Company B by N weeks, validated across N historical events) are automatically surfaced as "leading indicator" edge types. The canaries are encoded as graph edges with directional temporal lag metadata.

04 Relationship Decay Knowledge — "That partnership means less than it used to"

Weight Decay

Relationships change in strength over time, but databases don't update. A veteran analyst knows that a joint venture announced four years ago has become less operationally significant as one party diversified. They know that a supply chain relationship that was

critical in 2019 has been partially onshored by 2024. They know that executive relationships that drove a board interloc appointment are no longer active because the executives moved on. Capital IQ still shows all of these as active, with the same implied weight as day one. The analyst's calibrated decay knowledge is entirely unencoded.

> *"That JV is still in the database. But anyone who followed it knows the two companies barely interact anymore — one of them pivoted their strategy in 2022. The edge weight in any system treating it as 2019-equivalent is simply wrong."*

MKG BRIDGE POINT

Temporal edge weight decay: edges with no news reinforcement decay toward zero on a configurable schedule. WAN reads absence of co-mention, filing reference drops, and management departure signals to automatically reduce edge weights of aging relationships — encoding the decay knowledge that currently lives only in experienced analysts' heads.

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05 Sector Cycle Timing — "We're at week 6 of a 14-week pattern"

Cycle Timing

Experienced sector analysts develop precise mental models of cycle dynamics that go far beyond what any data system captures. They know the historical sequence of which companies announce first, how the market initially misreads the announcement, which second-order companies reprice with what typical delay, and how long the cycle typically lasts before a correction. This is Pattern Recognition built from witnessing 3–5 full cycles in a sector — typically 10–15 years of sector experience. No data system captures the cycle sequence as a structured, queryable, replayable pattern.

> *"I've seen four semiconductor upcycles. They all follow the same sequence. We're at week 6 of a typical 14-week propagation — TSMC has reported, ASML has confirmed the orders, now we're waiting for the memory players to inflect. The desk consensus is we're 40% through this particular cycle."*

MKG BRIDGE POINT

Temporal graph rewind enables cycle pattern comparison: replay graph state at any historical timestamp, diff two cycle starting points, and identify which edges activated in which sequence. Historical propagation sequences become queryable baselines for current cycle comparison — cycle timing knowledge becomes a graph query, not a PM's memory.

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06 Regulatory Co-Exposure — "Both face the same regulator, nobody is writing about it"

Regulatory

Certain companies face the same regulatory body, the same political risk exposure, or the same policy environment — and when scrutiny hits one, it systematically reaches the other. This knowledge is not in any commercial database. It comes from reading regulatory filings, understanding agency jurisdictions, following congressional committee assignments, and tracking which companies have shared lobbying relationships. An experienced DC-focused analyst knows which tech companies share a critical FTC commissioner relationship, which energy companies are on the same EPA permit renewal schedule, which banks have the same internal model validation cycle. This knowledge is almost never documented.

> *"Both these companies file under the same FERC docket number. Nobody outside DC knows this. When FERC ruled against the first one last month, the second one will follow within 90 days. I've watched this commissioner for six years."*

MKG BRIDGE POINT

Regulatory entity nodes in the graph: regulatory bodies, government agencies, and policy instruments become first-class entities with edges to the companies they govern. WAN reads regulatory news, agency announcements, and congressional filings to create co-exposure edges that are currently carried only in specialists' heads.

SECTION 03 · THE KNOWLEDGE DESTRUCTION EVENT

What a fund loses when a *15-year sector analyst walks out*

Bloomberg subscription: transferable. Factset model: exportable. Mental map of 200+ relationship edges, calibrated heuristics, cycle timing patterns, canary companies: completely lost. The talent drain is also a knowledge drain — and it happens every time.

✓ WHAT TRANSFERS (TOOL-CAPTURED)

- ✅ Bloomberg Terminal access — **$31,980/yr subscription**, fully re-assignable
- ✅ FactSet financial models — Excel exports, transferable to successor
- ✅ AlphaSense search history — can be reviewed and searched
- ✅ Written research memos and investment theses — archived
- ✅ Portfolio position history — logged in OMS/PMS systems
- ✅ Sell-side research saved to shared folders — retrievable

✗ WHAT IS PERMANENTLY LOST (TRIBAL KNOWLEDGE)

- ❌ Mental relationship map — **200+ edge weights calibrated from 15 years of sector observation**. AMAT lags ASML by 3 weeks. This substrate supplier is the canary for TSMC packaging. Lost.
- ❌ Company-specific calibration — this CFO sandbagged Q3 by 12% for 8 consecutive years. This one overstates margin expansion by a consistent factor. Lost.
- ❌ Cycle sequence knowledge — week-by-week propagation order across 4 historical semiconductor upcycles. Lost.
- ❌ Relationship decay calibrations — which JVs and partnerships have become operationally dormant despite active database entries. Lost.
- ❌ Regulatory co-exposure map — which companies share the same regulator, docket, or policy cycle. Never documented. Lost.

❌ Informal channel knowledge — which IR contacts tip off order flow changes ahead of announcements. Lost with their personal phone contacts.

THE MEASUREMENT

Investment management has a turnover-driven knowledge destruction cycle. Average PM tenure at a hedge fund is 3–5 years. Senior sector analyst tenure is longer but still ends. Every departure takes the bottom two layers of the iceberg with it. The top two (Bloomberg, FactSet) stay. The edge — the relationship map, the calibrations, the canaries — goes. The successor starts rebuilding from scratch, spending 12–18 months reconstructing what their predecessor had accumulated over a decade. This is a permanent, structural inefficiency at every fund.

SECTION 04 · THE RESEARCH WALKTHROUGH

Same event. Two analysts. *Radically different intelligence.*

TSMC reports earnings. A junior analyst and a 15-year semiconductor veteran sit at the same Bloomberg terminal. Here is exactly what each one does — and what tools cannot provide for either.

SCENARIO · TSMC Q4 EARNINGS RELEASE · JANUARY 2026 · 06:00 AM NY TIME

What each analyst layer does — and what no tool

BLOOMBERG / FACTSET

Both analysts open the TSMC earnings: Revenue $26.9B +37%, EPS $2.24 vs $2.10 est. AI server

Both see identical data.

TOOL COVERAGE: 100%

Fully automated. No edge for either analyst here.

	demand cited as primary driver. CoWoS packaging capacity mentioned.	
ALPHASENSE / RAVENPACK	Junior analyst searches AlphaSense for "TSMC CoWoS packaging" — finds 12 relevant analyst notes and 3 earnings call transcripts. Senior analyst also reads these, but with calibration: *she knows the CoWoS mention means the allocation increase to NVIDIA is larger than disclosed.*	TOOL COVERAGE: 60% (documents only) Junior gets the documents. Senior gets the implication. Gap = calibration layer.
TACIT — NO TOOL	Junior analyst notes NVIDIA, AMD, ASML as obvious beneficiaries. Senior analyst instantly mentally triggers her list of 14 second-, the memory company whose HBM allocation follows TSMC CoWoS by 6 weeks, and the Korean PCB maker whose order book including the substrate supplier no analyst covers	TOOL COVERAGE: 0% Junior misses 11 of 14 actionable names. All tribal knowledge. Zero tool coverage.

	order names —	changes with TSMC packaging capacity.	
TRIBAL — NO TOOL	Senior analyst shares with the desk: "This pattern is identical to Q2 2023. AMAT will follow in 17 days. The Taiwanese substrate play — that small-cap nobody covers — should move by week 3. We got burned missing that name last cycle."	Nobody writes this down. It's the desk's shared memory.	TOOL COVERAGE: 0% Verbal only. If this analyst leaves, this pattern recognition is gone. No audit trail.
MKG BRIDGE	MKG triggers automatically at T+60 seconds: graph traversal scores 47 entities. Surfaces the substrate supplier (score: 73), the HBM memory play (score: 68), the Korean PCB maker (score: 51). Shows lead-lag metadata:	The tribal knowledge is now encoded.	MKG COVERAGE: Encodes layer 3+4 The 15-year pattern is now a graph query. Survives analyst departure. No verbal required.

"This edge historically activates 17 days after TSMC CoWoS announcements (n=4 events, confidence: 78%)."

SECTION 05 · THE MKG BRIDGE POINTS

Precisely where MKG begins *to encode what was tribal*

MKG does not replace Bloomberg. It does not replace AlphaSense. It encodes what no tool has ever touched: the bottom two layers of the knowledge iceberg. Here are the precise bridge points.

TRIBAL KNOWLEDGE TYPE	BEFORE MKG (IN HEADS ONLY)	AFTER MKG (ENCODED IN GRAPH)
🗺️ **Secret Relationship Map**	"AMAT lags ASML by 3 weeks. The desk knows this. We watch it every cycle." Only the veteran knows the full 200-edge version of the map. Lost when she leaves.	Lead-lag edge metadata: temporal precedence relationships encoded from co-movement analysis across all historical events. The map becomes a queryable graph. The 200 edges survive the analyst.

TRIBAL KNOWLEDGE TYPE	BEFORE MKG (IN HEADS ONLY)	AFTER MKG (ENCODED IN GRAPH)
🎯 **CEO/CFO Calibration**	"This CFO guides down 12% every Q3, then beats by 10%. We just adjust. It's not written anywhere — it's just	Entity-level guidance bias model: historical guidance vs. actual delta per executive per quarter. Systematic over/under-guidance

	what everyone on the desk knows."	becomes a computable correction factor, not a verbal desk heuristic.

TRIBAL KNOWLEDGE TYPE	BEFORE MKG (IN HEADS ONLY)	AFTER MKG (ENCODED IN GRAPH)
🐦 **Canary Companies**	"There's a tiny Taiwanese substrate supplier that signals TSMC packaging capacity 6 weeks ahead. 10 analysts cover it globally. The desk watches it obsessively."	Leading indicator edge type: graph edges with temporal precedence metadata identify canary relationships automatically. Event history validates the lead-lag across N cycles. The canary becomes discoverable by any analyst, not just the veteran.

TRIBAL KNOWLEDGE TYPE	BEFORE MKG (IN HEADS ONLY)	AFTER MKG (ENCODED IN GRAPH)
📉 **Relationship Decay**	"That JV is still in the database, but the two companies barely interact anymore. Anyone treating it as 2019-equivalent is using a dead edge. The analyst who tracked the decay left two years ago."	Temporal weight decay: edges without recent news reinforcement automatically decay toward zero. Absence of co-mention, filing reference drops, executive departures = weight reduction. Dead edges become visible as dead, not phantom-active.

TRIBAL KNOWLEDGE TYPE	BEFORE MKG (IN HEADS ONLY)	AFTER MKG (ENCODED IN GRAPH)
🔄 **Cycle Sequence Timing**	"Week 6 of a typical 14-week propagation. We're at the ASML confirmation point. Memory players inflect next, then PCB suppliers, then software stack. I've seen four cycles.	Temporal graph rewind: historical propagation sequences stored and queryable. "Compare current state to Q2 2023 at T+6 weeks." Cycle positioning becomes a graph query, not a

	Everyone on the desk knows the sequence."	verbal consensus. New analysts access institutional cycle memory from day one.

TRIBAL KNOWLEDGE TYPE	BEFORE MKG (IN HEADS ONLY)	AFTER MKG (ENCODED IN GRAPH)
⚖️ **Regulatory Co-Exposure**	"Both companies file under the same FERC docket. Nobody outside DC tracks this. When the commission rules against one, the other follows within 90 days. Three times running. Nobody wrote it down."	Regulatory entity nodes: government agencies, commissions, and policy instruments are graph entities. Companies sharing a regulator are connected by a "regulatory co-exposure" edge. Ruling on one triggers graph traversal to all co-exposed entities. Never tribal again.

THE CORE INSIGHT

Bloomberg gives you what happened. AlphaSense gives you what was said. RavenPack gives you how people felt about it. FactSet gives you what it's worth.

But the $900K/year tool stack still cannot answer: which other companies does this signal reach, through what relationships, with what weights, in what sequence, with what historical precedent — in the next 60 seconds, automatically, with full explainability?

That answer has lived entirely in experienced humans' heads for 50 years. **MKG is the first system designed to encode it.**

SECTION 06 · COVERAGE MATRIX

What tools cover.
What MKG adds to it.

Knowledge Type	Bloomberg	AlphaSense	RavenPack	Capital IQ	FactSet
Real-time prices + data	✓	✗	✗	✗	~
News ingestion (multilingual)	~	✗	✓	✗	✗
Document search + NLP	✗	✓	~	✗	✗
Entity-level sentiment	✗	~	✓	✗	✗
Static relationship database	✗	✗	✗	✓	✗
Financial models + fundamentals	~	✗	✗	~	✓
● Dynamic edge weights	✗	✗	✗	✗	✗
● Lead-lag relationship metadata	✗	✗	✗	✗	✗
● CEO/CFO guidance calibration	✗	✗	✗	✗	✗
● Canary company identification	✗	✗	✗	✗	✗
● Relationship decay detection	✗	✗	✗	✗	✗
●● Historical cycle pattern storage	✗	✗	✗	✗	✗

Knowledge Type	Bloomberg	AlphaSense	RavenPack	Capital IQ	FactSet
●● Regulatory co-exposure edges	✗	✗	✗	✗	✗
●● Event propagation scoring	✗	✗	✗	✗	✗
●● Full signal explainability chain	✗	✗	✗	✗	✗

● Tacit knowledge layer · ●● Tribal knowledge layer · ✦ NEW = first commercial encoding of this capability

MKG SYSTEM · TRIBAL KNOWLEDGE ANALYSIS · FEBRUARY 2026 · CONFIDENTIAL

Next: Use Case Diagram →